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Investigating the association between clinical depression and medication adherence among working females aged 18-66 with Type 2 Diabetes in London, England

Research Methods
Research Question & Background Literature Review
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Background

Type 2 Diabetes (T2D) has emerged as a global health crisis, with its prevalence steadily increasing since the 1990s[1]. As of 2023, approximately 10% of the world's adult population (18+) was affected by this chronic metabolic disorder, characterized by high blood glucose levels resulting from insulin resistance or insufficient insulin production[1,2]. London represents a microcosm of this global challenge, with north of an estimated 7% of adults affected in 2021[3,4]. This concerning statistic, coupled with significant variation across boroughs, greater risk among diverse ethnic populations, and urban lifestyle factors such as poor diets and sedentary behaviours, marks the city as an area of notable concern[4,5].

While global T2D prevalence is generally higher in men, women face unique challenges that warrant specific attention[6]. Hormonal fluctuations, increased risks of cardiovascular disease (CVD), and a greater likelihood of mental-health issues, including depression, render women with T2D as a particularly vulnerable population[7]. Working females aged 18-66 represent an especially high-risk group, grappling with the complexities of disease-management alongside work-induced stress, family responsibilities, and societal pressures[8,9].

Clinical depression, 2-3 times more prevalent in individuals with diabetes than in the general population, can significantly impact self-management behaviours and overall health outcomes[10]. Despite medication adherence being a critical component of effective T2D-management, it is often compromised in patients with comorbid depression[11]. This poor adherence, left unaddressed, can lead to worsened glycaemic control, increased complications, and higher mortality rates[12,13].

Understanding the interplay between clinical depression and medication adherence in working-age women with T2D is crucial for developing targeted interventions and improving public health outcomes. By addressing barriers to adherence, healthcare providers and policymakers can implement more effective strategies for T2D-management in this demographic, potentially mitigating long-term health and economic impacts of this growing global health concern.

Literature Review

Depression and T2D

The association between depression and T2D has been well-documented. Studies indicate that individuals with T2D are significantly more likely to experience clinical depression compared to their non-diabetic counterparts[10,14,15]. Meta-analyses, such as that conducted by Lustman *et al.*, reveal that depression exacerbates the progression of diabetes through physiological mechanisms, including inflammation and dysregulation of stress hormones, negatively impacting glucose control[11]. Conversely, the stress associated with managing chronic conditions like T2D increases the risk of developing depression[16]. This bi-directional relationship underscores the need for approaches addressing both conditions simultaneously. For working women, these challenges are compounded by the interplay of professional, caregiving, and personal responsibilities, which heighten psychological stress and hinder effective disease-management[9,17–19].

Psychological Stress and Disease Management

Research by Mendes *et al.* emphasizes the heightened stress faced by this female demographic, amplifying depressive symptoms and complicating T2D-managment[19]. Despite growing recognition of this issue, limited research explores these dynamics in urban contexts like London's, leaving a gap in the literature. What is clear, is that depression is a major predictor of poor medication adherence across chronic conditions, including diabetes. Meta-analysis of 31 studies by Grenard *et al.* estimates depressed individuals have 1.76 times the odds of non-adherence compared to their non-depressed counterparts, owing primarily to cognitive impairments, low motivation, and feelings of hopelessness[20]. Earlier meta-analyses, such as those by DiMatteo *et al.* and Gonzalez *et al.*, have estimated even higher odds, up to three-fold[12,21]. For T2D patients, poor adherence leads to inadequate glycaemic control, increasing the risk of complications such as CVD and retinopathy[22].

The London Context

While depression is recognized as a risk factor for T2D, its role in mediating non-adherence remains unclear, indicating a critical area for further research. Investigating this association is association is particularly urgent for working women, whose stressors are often intensified by rigid work schedules, limited access to flexible health services, and inadequate social support networks[9,18]. Cultural and socio-economic factors unique to diverse populations concurrently influence adherence behaviours, requiring localized and nuanced approaches to intervention design[23]. Workplace wellness programs and sex-sensitive healthcare policies could represent promising avenues, especially if informed by robust evidence specific to London's working female population.

Theoretical Frameworks Informing Adherence Behaviours

Theoretical frameworks such as Social Cognitive Theory provide valuable insights into how depression might affect adherence behaviours[24]. In T2D, depression can diminish self-efficacy around medication adherence, leading to disengagement from vital disease management practices. The Stress Coping Model posits that chronic stress, such as that experienced by working women balancing roles, can foster maladaptive behaviours and deplete coping mechanisms[25]. These frameworks suggest enhancing self-efficacy and reducing stress could potentially improve adherence outcomes.

Methodological Limitations in Existing Research

Existing research presents limitations that hinder its applicability to working women in London. While the literature provides fundamental insights into the complex relationships between depression, T2D, and medication adherence, it fails to cover how these factors specifically affect these women in London's unique socio-economic and multi-cultural context. Characterized by its diversity, disparity, and collectivist healthcare system, the city necessitates localized research to inform effective workplace strategies and policies for such a vulnerable population.

Despite the strong internal validity offered by randomized controlled trials (RCTs), ethical and logical constraints, coupled with limited external validity, suggest observational designs are better suited to capture personal behaviours of adherence. Meta-analyses in existing literature relied on such designs, enabling for extensive study of these sensitive behaviours. However, most outlined inclusion of U.S.-based studies, reflecting cultural and systemic differences that limit generalizability elsewhere. Generalized findings across the many diverse American subpopulations overlook the unique intersections of occupation, sex, and social factors affecting medication adherence in London.

Reliance on self-reported adherence data also raises concerns around validity due to potential for social-desirability and recall bias, skewing calculated estimates. Mixed-methods cohort studies that consider these factors and allow for temporal assessments of sensitive behaviours are essential for developing effective, targeted interventions in London. These could employ probability-based stratified sampling and quantitative positivism to assess the association between depression and medication adherence, and qualitative relativism to explore lived experiences of barriers, informing a pragmatic approach.

Rationale

- There is a critical lack of studies addressing the association between clinical depression and medication adherence among working females with T2D in London, a group uniquely affected by overlapping stressors of professional, personal, and caregiving responsibilities which impact health behaviours.
- Understanding this bi-directional relationship is essential for designing targeted, context-specific interventions that improve medication adherence and health outcomes for this vulnerable demographic.
- Investigating this relationship within London's diverse socio-economic and cultural context can provide valuable insights to inform local policies and workplace wellness strategies, enhancing support for working women managing chronic conditions.

Research Question

Literature review therefore begs the question: *"What is the association between clinical depression and medication adherence among working females (18-66 years) with Type 2 Diabetes in London, England?"*

Reflection Statement

The literature review process revealed valuable insights and significant research gaps, despite an abundance of existing studies. Quality varied due to inconsistencies in study designs, definitions of medication adherence, and tools of measurement. Access to peer-reviewed studies through academic databases was straightforward, complemented by grey literature for local context. Key sources included robust meta-analyses, reputable journals like *Diabetes Care*, and work by other influential organizations such as the American Diabetes Association. However, geographic bias towards the U.S. underscored the need for UK-centric research. Future studies should consider dynamics between biological, psychological, and social dimensions in T2D-Management to better investigate medication adherence within London's socio-cultural and healthcare landscape.

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